

Queue

1. Queue in Data Structure is -----

- a) FIFO
- b) LIFO
- c) Ordered array
- d) Linear tree

Answer: a

2. Process of inserting an element in queue is called -----

- a) Create
- b) Push
- c) Enqueue
- d) Dequeue

Answer: c

3. Process of inserting an element in queue is called -----

- a) Enqueue
- b) Push
- c) Evaluation
- d) Dequeue

Answer: d

4. A linear list of elements in which deletion can be done from one end (front) and insertion can take place only at the other end (rear) is known as -----

- a) Queue
- b) Stack
- c) Tree
- d) Linked list

Answer: a

5. The data structure required for Breadth First Traversal on a graph is?

- a) Stack
- b) Array
- c) Queue
- d) Tree

Answer: c

6. Circular Queue is also known as -----

- a) Ring Buffer
- b) Square Buffer
- c) Rectangle Buffer

d) Curve Buffer

Answer: a

7. If the elements “A”, “B”, “C” and “D” are placed in a queue and are deleted one at a time, in what order will they be removed?

a) ABCD

b) DCBA

c) DCAB

d) ABDC

Answer: a

8. A data structure in which elements can be inserted or deleted at/from both ends but not in the middle is?

a) Queue

b) Circular queue

c) Dequeue

d) Priority queue

Answer: c

9. A normal queue, if implemented using an array of size MAX_SIZE, gets full when?

a) $\text{Rear} = \text{MAX_SIZE} - 1$

b) $\text{Front} = (\text{rear} + 1) \% \text{MAX_SIZE}$

c) $\text{Front} = \text{rear} + 1$

d) $\text{Rear} = \text{front}$

Answer: a

10. If the MAX_SIZE is the size of the array used in the implementation of circular queue. How is rear manipulated while inserting an element in the queue?

a) $\text{rear} = (\text{rear} \% 1) + \text{MAX_SIZE}$

b) $\text{rear} = \text{rear} \% (\text{MAX_SIZE} + 1)$

c) $\text{rear} = (\text{rear} + 1) \% \text{MAX_SIZE}$

d) $\text{rear} = \text{rear} + (1 \% \text{MAX_SIZE})$

Answer: c

11. If the MAX_SIZE is the size of the array used in the implementation of circular queue, array index starts with 0, front point to the first element in the queue, and rear point to the last element in the queue. Which of the following condition specify that circular queue is FULL?

a) $\text{Front} = \text{rear} = -1$

b) $\text{Front} = (\text{rear} + 1) \% \text{MAX_SIZE}$

c) $\text{Rear} = \text{front} + 1$

d) $\text{Rear} = (\text{front} + 1) \% \text{MAX_SIZE}$

Answer: b

12. If the MAX_SIZE is the size of the array used in the implementation of circular queue, array index starts with 0, front point to the first element in the queue, and rear point to the last element in the queue. Which of the following condition specify that circular queue is EMPTY?

- a) Front = rear = 0
- b) Front = rear = -1
- c) Front = rear + 1
- d) Front = (rear + 1) % MAX_SIZE

Answer: b

13. Queues serve major role in -----

- a) Simulation of recursion
- b) Simulation of arbitrary linked list
- c) Simulation of limited resource allocation
- d) Simulation of heap sort

Answer: c

14. Which of the following is not the type of queue?

- a) Ordinary queue
- b) Single ended queue
- c) Circular queue
- d) Priority queue

Answer: b

15. Following is C like pseudo code of a function that takes a Queue as an argument and uses a stack S to do processing.

```
void fun(Queue *Q)
{
    Stack S;
    while (!isEmpty(Q))
    {
        push(&S, deQueue(Q));
    }
    while (!isEmpty(&S))
    {
        enqueue(Q, pop(&S));
    }
}
```

What does the above function do in general?

- a) Remove the last from Q
- b) Keeps the Q same as it was before the call
- c) Makes Q Empty

d) Reverses the Q

Answer: d

16. Which one of the following is an application of Queue Data Structure?

- a) When a resource is shared among multiple consumers.
- b) When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes
- c) Load Balancing
- d) All of the above

Answer: d

17. How many stacks are needed to implement a queue? Consider the situation where no other data structure like arrays, linked list is available to you.

- a) 1
- b) 2
- c) 3
- d) 4

Answer: b

18. How many queues are needed to implement a stack? Consider the situation where no other data structure like arrays, linked list is available to you.

- a) 1
- b) 2
- c) 3
- d) 4

Answer: b

19. A circular queue is implemented using an array of size 10. The array index starts with 0, front is 6, and rear is 9. The insertion of next element takes place at the array index.

- a) 0
- b) 7
- c) 9
- d) 10

Answer: a

20. In linked list implementation of a queue, front and rear pointers are tracked. Which of these pointers will change during an insertion into a NONEMPTY queue?

- a) Only front pointer
- b) Only rear pointer
- c) Both front & rear pointer
- d) None of the front & rear pointer

Answer: b

21. In linked list implementation of a queue, front and rear pointers are tracked. Which of these pointers will change during an insertion into EMPTY queue?

- a) Only front pointer

- b) Only rear pointer
- c) Both front & rear pointer
- d) None

Answer: c

22. An array of size MAX_SIZE is used to implement a circular queue. Front, Rear, and count are tracked. Suppose front is 0 and rear is MAX_SIZE -1. How many elements are present in the queue?

- a) Zero
- b) One
- c) MAX_SIZE - 1
- d) MAX_SIZE

Answer: d